Compact and Fast ADCs for Pixel Detectors

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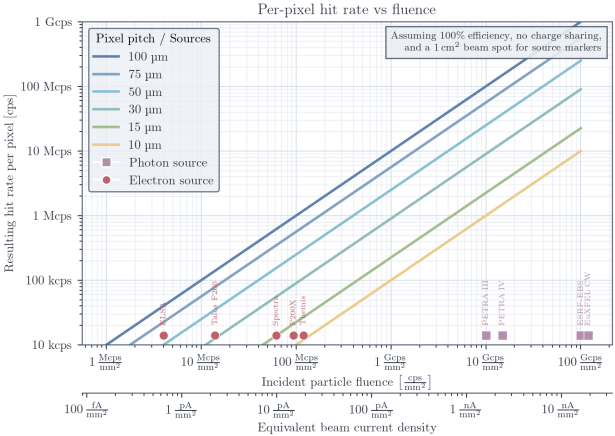
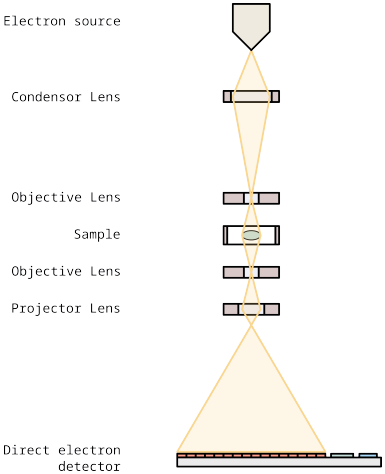
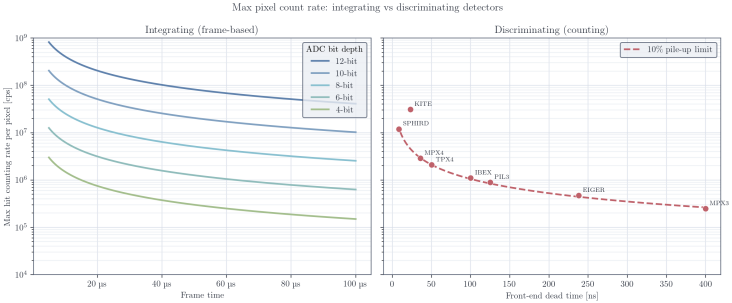
DPG 2026

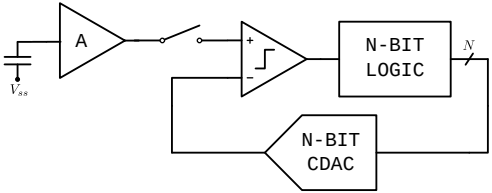
Pixel Detectors & Hit Rates

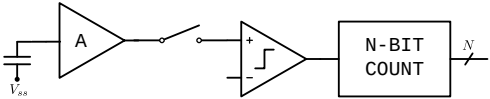




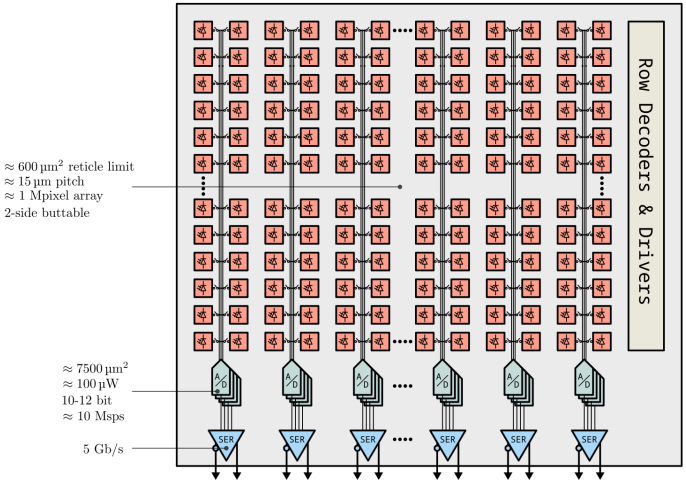
Integrating vs Discriminating Detectors



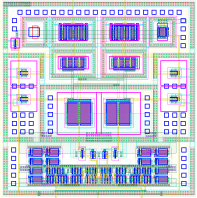




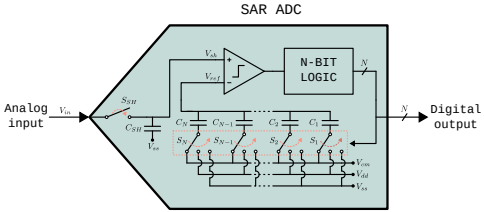
System Architecture

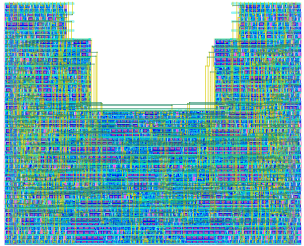


SAR ADC Architecture

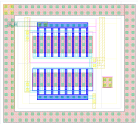


Comparator:
- $<100\ \mu V_{rms}$ noise
- $<20\ \mu W$ power
- <2 ns metastability

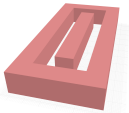




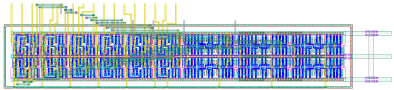
SA Digital Logic:
- 17-bit state machine
- Manual bit ctrl for testing
- Programmable init. state



Samp. Switch:
- 200 MHz BW
- No bootstrap
- $<1\ k\Omega\ R_{on}$

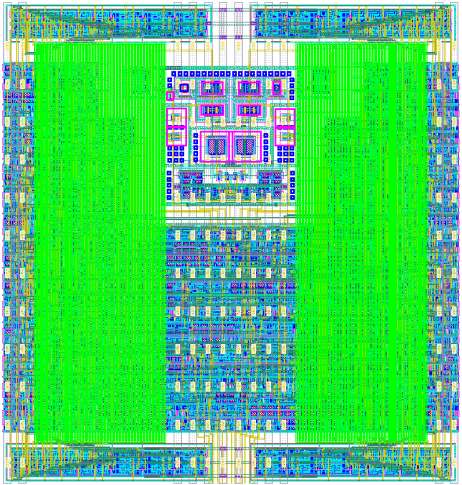


CDAC:
- 1.6 pF total
- 0.4 aF unit
- M5–M7 fringe

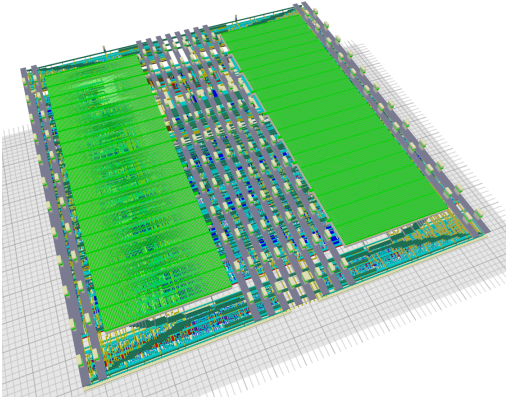


Cap Drivers:
- 16-bit, scaled to caps
- $\sim$1 ns settling

Full ADC Layout



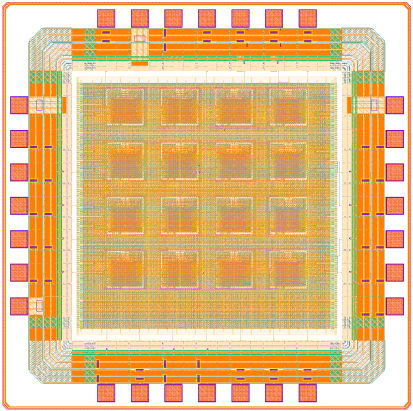
2D: $60 \times 60\ \mu\text{m}$



3D view with CDAC arrays

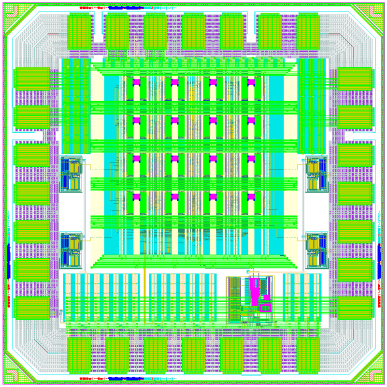
Prototype Chips

Initially designed w/ FOSS PDK & tools:

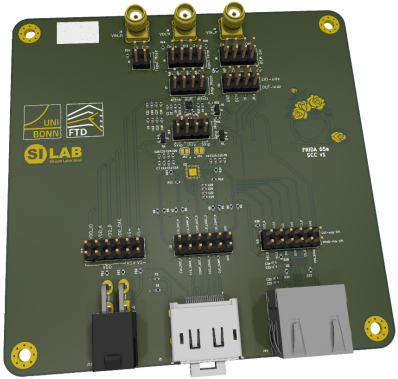


IHP SG13G2 (130 nm) — 1.6×1.6 mm

Ported for final submission:



TSMC 65 nm LP — 1×1 mm

Outlook

100 test ASICs received, ready for wirebonding on carrier board